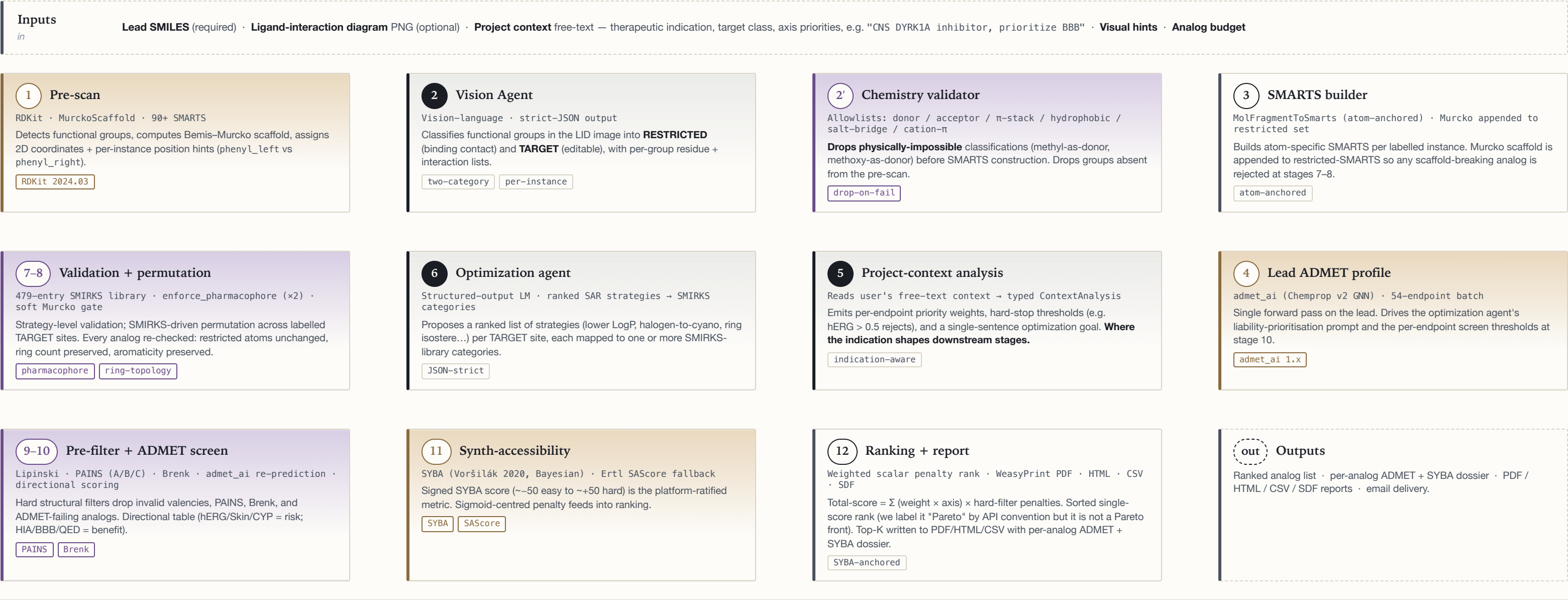


FIGURE 1

A vision-language agentic pipeline for lead optimization

Twelve deterministic stages map a lead structure, its ligand-interaction diagram, and a free-text project-context input to a ranked set of synthetically-accessible analogs. Agent stages (dark fill) are wrapped by deterministic gates (purple) so a perception error in stage 2 cannot corrupt the analog set downstream. Pipeline reads as a serpentine: row 1 left→right, row 2 right→left, row 3 left→right.



STAGE COLOUR

- Agent (language model)
- Open-source substrate
- Defence-in-depth gate
- Pipeline logic

SUBSTRATE CITATIONS

RDKit Landrum et al., 2024.03 · **admet_ai**/Chemprow v2 Swanson et al., 2024 · **SYBA** Voršilák et al., 2020 (AUC > 0.81) · **SAScore** Ertl & Schuffenhauer, 2009 · **PAINS** Baell & Holloway, 2010 · **Brenk** Brenk et al., 2008 · **Bemis–Murcko** Bemis & Murcko, 1996.

¹ **Per-instance labelling.** Two phenyls in the same molecule would collapse to a single generic SMARTS (c1ccccc1) without disambiguation. The pipeline derives left/right/top/bottom hints from RDKit 2D coords and emits atom-anchored SMARTS via `MolFragmentToSmarts`, so each instance carries a distinct restricted/editable verdict.

² **Soft Murcko gate.** The Bemis–Murcko scaffold SMARTS is appended to the restricted set at stage 3. The gate itself is implemented by `enforce_pharmacophore()`, called twice — at strategy validation (one example per SMIRKS) and during permutation (every candidate analog). Predicates: restricted atoms unchanged, ring count preserved, aromaticity preserved.

³ **Chemistry validator coverage.** Six allowlist tables (H-bond donor / acceptor / π -stack / hydrophobic / salt-bridge / cation- π) map each functional-group family to its allowed interactions. Vision output that violates the chemistry tables is dropped before stage 3.

⁴ **SMIRKS library scope.** 479 curated SMIRKS transformations across 22 categories (aromatic-ring swaps, amine modifications, carbonyl modifications, CNS penetration, nitrogen-heterocycle swaps, polarity adjustments, halogen substitutions, etc.). All entries carry `validated=True`, but validation is curator-asserted and not yet gated by an automated CI test — known limitation.

⁵ **Ranking metric.** The `total_score` is a weighted scalar sum across docking, ADMET liabilities, SYBA penalty, and structural alerts, multiplied by hard-filter penalties. Output is single-sorted; the field label "Pareto" is API-convention, not a real Pareto front. Honest framing: *weighted scalar penalty rank with SYBA as the synth-accessibility anchor*.

⁶ **Project context.** A free-text input — for example "CNS kinase, prioritize BBB" or "oral gram-positive antibacterial". Stage 5's language model parses this into a typed `ContextAnalysis` record and writes its weights into the ranker and the ADMET-screen thresholds. Without context, the pipeline falls back to a generic balanced default.